Formal Proof of 'Living Expression' and 'Self-Consistent Computability' in Open Complex Giant Systems

Formal Verification Based on Dual-Helix 5D Manifold and Meta-Cognitive Re-Bounding

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"Any computability and optimal solution strictly depend on the observer's 'boundary drawing'. Inside the boundary, if conditions are self-consistent, the system is closed-loop computable and converges to an in-boundary relative optimal solution S_k^*."
"Open systems continuously exchange matter, energy, and information with the environment. By Gödel's Incompleteness Theorem, finite boundary drawing is inevitably breached by new residuals. The essence of a living system is to incorporate new perturbations via endogenous 'meta-cognition' to complete dynamic re-bounding (Boundary$_k$ → Boundary$_{k+1}$), perpetually sustaining self-consistency."
"Living is absolute; optimality is relative inside boundaries. Self-consistency ≡ in-boundary closed-loop computability; living ≡ meta-cognition-driven continuous re-bounding."

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1. Epistemological Principle 1: Boundary Medium and 5D Manifold Minimal Completeness

1.1 Boundary Drawing is the Precondition for Computability

Physical reality is in-accessible directly; an observer must draw boundaries to establish an Ought-to-be System (the observer-constructed model). Eliminating work vector cancellation ($\boldsymbol{v}_i, \boldsymbol{v}_j$) inside boundaries achieves conditional self-consistency:

$$\text{Physical Reality (Strongly Coupled)} \xrightarrow{\text{Boundary Drawing}} \text{Ought-to-be Model (Self-Consistent Medium, } \mathcal{M}_{\text{model}}) \\ \Longrightarrow \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0 \Longrightarrow \Delta W_{\text{heat}} \to 0$$

1.2 Theorem 1.1 (Minimal Completeness Theorem of 5D Manifold)

The observer's boundary-drawing action (Boundary$_k$) necessarily and uniquely derives the Ought-to-be topological manifold:

$$\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$$

The manifold presents a dual-layer nested topology: "Rigid Structural Base Manifold $\langle \mathcal{N}, \mathcal{T}, \mathcal{C} \rangle$ + Dynamic Causal Tangent Fibers $\langle X, \Delta X \rangle$".

- **Completeness (Stripping one dimension causes structural collapse)**:
  - Stripping N (Node) → **Nothingness** (spatial entity measure zero, system reduced to void);

    - Stripping T (Topology) → **Disorder** (latent dependency network broken, elements degraded to isolated sand);
    - Stripping C (Constraint) → **Unboundness** (exclusivity and conservation boundary lost, feasible region divergence causing factorial explosion);
    - Stripping X (State Vector) → **Statelessness** (potential energy distribution configuration lost, system losing spatiotemporal calibration);
    - Stripping ΔX (State Transition) → **Causallessness / Feedbackless** (state transition pedigree severed, who transforms into whom unknown, causal chain broken, residual with causality unable to compare, closed loop invalidated).
- **Minimality (Algebraic independence)**: All 5 dimensions are mutually orthogonal and independent; no single dimension can be derived from the remaining four.

Conclusion: Completeness + Minimality $\Longrightarrow \mathcal{M}_{\text{model}}$ is the minimal complete basis of the Ought-to-be System (the observer-constructed model). Q.E.D.

[Algebraic Master Equation of the Latter Two Dimensions and Causal-Residual Dual Transformation Law]:
The latter two dimensions dynamically close via the differential algebraic master equation $X_{\text{plan}} = X \oplus \Delta X_{\text{causal}}$:

1. **Forward Causal Generation**: The forward deduction side ΔX carries the **instantiated causal chain (who transforms into whom, such as causal work in phase transition from water to ice)** locked from preceding states to subsequent targets, deriving next-moment planned state X_{plan};
2. **Reverse Residual Manifestation**: After real measured state X_{real} manifests, shifting terms in the master equation yields **[residual with causality]** $\Delta X_{\text{residual}} = X_{\text{real}} \ominus X_{\text{plan}}$;
3. **Traceability Judgment along Causal Chain**: Reverse-trace the residual along the original causal chain—if the causal chain holds and only physical entities are obstructed, judge "the world is off", issuing control work to govern the world; if the causal chain breaks and the feasible set is empty, judge "the model is wrong", activating second-order meta-cognitive operator Φ for constitutional amendment and model reconstruction. If the world is off, adjust execution; if the model is wrong, apply second-order amendment.

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2. Dual-Helix Structure $S = \mathcal{M}_{\text{model}} \otimes \mathcal{A}$ and Its Physical Propositions

The holistic representation of the system is defined as the tensor product of topological manifold $\mathcal{M}_{\text{model}}$ ("Form") and forward causal evolution operator $\mathcal{A} = \bigodot_{k=1}^{K}(\mathcal{B}_k \circ \mathcal{P}_k)$ ("Function"):

$$S = \mathcal{M}_{\text{model}} \otimes \mathcal{A}$$

2.1 Right Helix (Function): Theorem 2.1 (Closed-Loop Computability Theorem)

Note: Theorem 2.1 / Proposition 2.2 in this paper are consistent in conclusion with "Causal Finiteness Lemma 1.1" in the Order monograph and Theorem 7.2 / Proposition 7.3 in The Second Renaissance; the numbering differences arise because each paper is written independently.

Physical Proof (Discrete Clock & Bounded Relaxation $\Longrightarrow$ Polynomial Closed-Loop Computability)

1. Boundary drawing constrains phase space to finite entity scale ($N < +\infty$) over a bounded-degree causal DAG ($|\mathcal{E}| = \mathcal{O}(N)$ via shared-node merging);

2. The control system's clock step $\Delta t_{\text{clock}} > 0$ and physical instability relaxation window $\tau_{\text{phy}} < +\infty$ strictly truncate the maximum DAG depth to a finite constant: $K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}} < +\infty$;

3. Orthogonally separating accumulable extensive flows ($\mathcal{O}(N)$ kitting pull) from perishable intensive capacities ($\mathcal{O}(N\log N)$ local window sorting) eliminates cross-stage backtracking—because kitted extensive flows persist across $t' \geq t_{\text{avail}}$—ensuring single-step deterministic closure under $\mathcal{B}_k \circ \mathcal{P}_k$ in $\mathcal{O}(N\log N)$.
Conclusion: Finite Entity Scale (N) + Finite Causal Depth ($K \leq \frac{\tau_{\text{phy}}}{\Delta t_{\text{clock}}}$) + Polynomial Non-Backtracking Operators ($\mathcal{P}_k \sim \mathcal{O}(N\log N)$) $\Longrightarrow$ Closed-loop computation time $T(N) = \mathcal{O}(K \cdot N\log N) \in \boldsymbol{P}$, with $T(N) \leq \tau_{\text{phy}} < +\infty$. The system strictly completes self-consistent closed-loop computation before physical instability occurs. Q.E.D.

2.2 Left Helix (Form): Proposition 2.2 (Constraint Walk-Through, Zero Cancellation, and Unique In-Boundary Optimality)

Within the Dual-Helix framework, global in-boundary optimality and trajectory uniqueness ($\exists! S_k^*$) follow directly from physical loss decomposition and the vector triangle inequality:

[Proposition 2.2 (Constraint Walk-Through ∧ Zero Cancellation ⇔ Unique In-Boundary Optimality $\exists! S_k^*$)]
Within a given boundary Boundary_k, the constraint manifold $C_k = \langle C_{\text{core}}, C_{\text{param}} \rangle$ encodes hard physical conservation C_{core} (spatiotemporal exclusivity and flux conservation) and target work commitments C_{param} (excluding trivial idle zero-work $\boldsymbol{v}_i = \mathbf{0}$). The evolution algorithm $\mathcal{A}$ permits only co-directional steps ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$), while pruning operator $\Pi_\perp$ rigidly strips non-orthogonal conflicting branches. Any walked-through trajectory satisfies the physical equivalence:

$$\begin{aligned} &\text{Constraint Walk-Through } S_k^* \ \wedge \ (\mathcal{W}_{\text{heat}}[S_k^*] = 0) \\ \Leftrightarrow\ &\text{Unique In-Boundary Optimal Trajectory } (\exists! S_k^* \\ =\ &\arg \min_{S \in \Omega_{\text{feasible}}(C_k)} W_{\text{loss}}^{\text{total}}(S)) \end{aligned}$$

Formal Physical Proof:

1. **Direct-Sum Loss Decomposition**: For any admissible trajectory $S \in \Omega_{\text{feasible}}(C_k)$ satisfying both C_{core} and C_{param}, total system loss decomposes into $W_{\text{loss}}^{\text{total}}(S) = W_{\text{intrinsic}}(C_{\text{core}}) + \mathcal{W}_{\text{heat}}[S]$, where $W_{\text{intrinsic}}(C_{\text{core}})$ is the irreducible physical floor dictated by hard conservation limits and $\mathcal{W}_{\text{heat}}[S] \geq 0$ is internal vector-cancellation waste heat;

2. **Global Optimality by Contradiction**: Advancing along causal characteristic lines under $\Pi_\perp$ truncates normal shear components ($\boldsymbol{v}_\perp = \mathbf{0}$), ensuring $\delta W_{\text{heat}}^{(k)} = 0$, $\mathcal{W}_{\text{heat}}[S_k^*] = 0$, and $W_{\text{loss}}^{\text{total}}(S_k^*) = W_{\text{intrinsic}}(C_{\text{core}})$. Suppose by contradiction that another walked-through trajectory $S' \in \Omega_{\text{feasible}}(C_k)$ achieved strictly lower loss $W_{\text{loss}}^{\text{total}}(S') < W_{\text{loss}}^{\text{total}}(S_k^*)$; because $W_{\text{intrinsic}}(C_{\text{core}})$ is invariant under fixed C_{core}, this would require $\mathcal{W}_{\text{heat}}[S'] < 0$, contradicting the Euclidean vector triangle inequality $\mathcal{W}_{\text{heat}}[S] \geq 0$;

3. **Uniqueness ($\exists! S_k^*$)**: Combining spatiotemporal resource exclusivity with a strict lexicographic tie-breaking order $(t_{\text{avail}}, T_{\text{commit}}, -\mathcal{P}, \text{ID})$ for simultaneous equal-urgency tasks eliminates symmetric permutation degeneracy, rendering each step deterministic and guaranteeing a unique optimal trajectory $\exists! S_k^*$. Q.E.D.

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3. Gödelian Incompleteness and Meta-Cognitive Dynamic Re-Bounding

- **Gödelian Residual**: Open complex giant systems continuously exchange matter, energy, and information with the environment. By Gödel's Incompleteness Theorem, any finite boundary drawing Boundary_k cannot envelope external continuous perturbations, inevitably manifesting cotangent residuals exceeding internal computation: $\| \Delta X(t) \| > \theta$.
- **Meta-Cognitive Amendment Operator** Φ: When residuals breach the tolerance threshold, the system activates meta-cognitive operator Φ: $\text{Boundary}_k \xrightarrow{\Delta X} \text{Boundary}_{k+1}$:
    1. Keeping the immutable physical core C_{core} invariant;
    2. Relaxing secondary rules C_{sec};
    3. Absorbing new residuals into constraint set C_{k+1}.
       If self-consistency ($\text{Sol} \neq \emptyset, \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$) is restored within relaxation window τ_{phy}, the system completes a phase transition / generational constitutional amendment leap; otherwise, it degrades, collapses, and halts (system halting/death).

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4. Three-Tier Spatiotemporal Unification Theorem of Residuals for Living Closed Loop

Tier	Physical Dimension	Core Control Law	Physical & Systems Science Verdict
Tier 1	**Spatial Domain**	Eliminate vector cancellation ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$)	Eliminates friction waste heat, guaranteeing structural integrity
Tier 2	**Temporal Domain**	DAG depth physical truncation ($K < +\infty$)	Locks polynomial complexity, guaranteeing computation within time bounds
Tier 3	**Residual Domain**	Gödel residual triggers meta-cognitive operator Φ	Absorbs external perturbations, guaranteeing non-rigidification of paradigms

Living Governance Three-State Loop:

Generation (Boundary Est.) → Survival (Self-Consistent Computing)
→ Evolution (Meta-Cognitive Re-bounding)

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5. Popperian Falsifiability Matrix

To defend the academic dignity of this framework as hard-core meta-science, all core propositions are provided with explicit, independently verifiable counter-factual conditions for falsification:

Core Theorem / Proposition	Observational Metric	Counter-Factual Condition for Falsification
Theorem 1.1 (5D Manifold Completeness)	Phase space measure & topological basis	Discovery of any state transition unrepresentable by 5D; or proof that 1 dimension is linearly expressible by the other four.
Theorem 2.1 (Closed-Loop Computability)	Computation time $T(N)$ scaling curve	Under bounded causal depth K, algorithm exhibits unavoidable exponential explosion $O(2^N)$ or halting deadlocks.
Proposition 2.2 (Traversability ≡ Zero Cancellation ≡ Optimal)	Pruning trajectory & friction functional	Algorithm pruning fails; traversed path exhibits normal shear components; or a lower-friction valid trajectory exists.
Meta-Cognitive Operator Φ	Residual magnitude $\| \Delta X(t) \|$ & structural re-bounding	External unknown residuals exceed threshold θ long-term, yet system neither re-bounds nor collapses and survives forever.

6. Conclusion: Living Governance Paradigm and Cross-Domain Inheritance from a Meta-Science Perspective

1. **Boundary Relativity Law**: No "absolute universal optimal solution" exists outside boundaries; any optimal solution S_k^* is strictly relative to its boundary Boundary_k; **Living is absolute (referring to closed-loop survival); optimality is relative inside boundaries.**
2. **Self-Consistent Computability Law**: In-boundary conditional self-consistency ($\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle \geq 0$) and finite causal steps ($K < +\infty$) are necessary and sufficient physical conditions for polynomial closed-loop computability.
3. **Meta-Cognitive Evolution Law**: Gödelian residuals ($\| \Delta X \| > \theta$) force second-order meta-cognitive operator Φ to dynamically re-bound ($\text{Boundary}_k \rightarrow \text{Boundary}_{k+1}$), establishing structural isomorphism across cybernetics, statistical physics, computational graph theory, and biological evolution.
4. **Two-Layer Structural Isomorphism Across Domains**: Because Theorem 2.1 and Proposition 2.2 depend solely on three domain-invariant physical properties—(i) bounded-DAG multi-prerequisite kitting gates, (ii) forward persistence of accumulable extensive flows versus spatiotemporal exclusivity of perishable intensive capacities, and (iii) discrete clock truncation ($K < +\infty$)—any Open Complex Giant System (whether global manufacturing networks, cloud AI compute clusters, smart power grids, hospital surgical centers, or Multi-Agent systems) mapped onto $\mathcal{M}_{\text{model}} = \langle \mathcal{N}, \mathcal{T}, \mathcal{C}, X, \Delta X \rangle$ **isomorphically inherits polynomial closed-loop computability ($T \in \boldsymbol{P}$) and unique in-boundary orthogonal optimality ($\exists! S_k^*$)**.

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